COVID RESILIENCE

Who coped best with COVID-19?

Six criteria, open data from OWID, the World Bank, the IMF and Oxford, and one transparent weighted score

AI-GENERATED Analysis, code and slides generated by Claude (Anthropic AI)

THE GOAL

“Coped best” isn’t a single number

The project started as a search for the best vaccine. That stalled: vaccine-brand data exists for only 44 locations, against 255 in the main dataset.

So the question became broader: **which countries came through the pandemic best?**

What we want to deliver

- An explicit definition of resilience
- Comparable figures per country, from citable open data
- A ranking whose weights you can see and change
- A check of how robust that ranking is

HOW WE DEFINE RESILIENCE

Six criteria — lower is better on each

Health

Excess deaths per million (or reported deaths, or the life-expectancy drop)

Economy

GDP loss 2020–23 vs. the IMF's pre-COVID forecast (or the 2015–19 trend)

Jobs

Rise in unemployment: highest of 2020–22 minus 2019

Education

Weeks that schools were fully closed, 2020–22

Freedom

Average stringency of government measures, 2020–22

Recovery

Years that GDP stayed below its 2019 level

Health is the core criterion (orange). Pre-COVID healthcare capacity is shown as an explanation, not scored.

HOW WE DEFINE RESILIENCE

Four choices that change the ranking

1 · The end date

New Zealand and Australia shine until 2021, then the virus arrived. We measure up to the end of 2023.

2 · Like with like

Islands and young countries start ahead. Compare Belgium with its neighbours, not with New Zealand.

3 · Data quality

Countries with poor death registration are missing, or look better than they were.

4 · The weights

The weights largely decide the winner, so they are explicit, adjustable and stress-tested.

Each choice is a control in the app: end-date slider, peer group, data-quality tab, weight sliders

PART 2 · THE DATA

Six open sources, one SQLite database, one row per country

Every number in the app can be traced back to a public download, and every calculated value is marked as derived.

WHERE THE DATA COMES FROM

The sources at a glance

Source	Grain	Used for
Our World in Data (OWID)	daily	Excess & reported deaths, stringency, vaccination, ICU
Oxford policy tracker (OxCGRT)	daily	School closures (indicator C1M)
World Bank — GDP	yearly	Real GDP and growth, 2010–2025
World Bank — social	yearly	Unemployment, life expectancy
World Bank — health	yearly	Health spending, doctors, nurses, beds
IMF WEO, October 2019	yearly	Pre-COVID GDP forecast: the “no COVID” baseline

Full URLs and retrieval dates: sources.md · retrieved Aug–Nov 2026

SOURCE 1 · OUR WORLD IN DATA

The backbone: daily health data per country

429,436

rows in owid-covid-data.csv

239

locations kept after filtering

Column we use	Upstream source
excess_mortality_cumulative_per_million	Human Mortality DB, World Mortality Dataset
total_deaths_per_million	WHO COVID-19 dashboard
stringency_index	Oxford OxCGRT, 2020–2022
people_fully_vaccinated_per_hundred	National vaccination reports
icu_patients_per_million	National reports, mostly richer countries

github.com/owid/covid-19-data · retrieved 2026-08-24

SOURCES 2 & 3 · WORLD BANK AND IMF

Yearly figures for economy, jobs and health care

World Bank API

World Development Indicators, 2010–2025, downloaded as JSON:

- Real GDP (constant 2015 US$) and growth
- Unemployment (ILO estimate)
- Life expectancy at birth
- Health spending, physicians, nurses, hospital beds

Country metadata flags aggregates such as “European Union” so they can be dropped.

IMF World Economic Outlook, Oct 2019

Published **before anyone knew about COVID**. Its growth forecasts for 2020–2024 tell us what each economy was expected to do without a pandemic.

194 countries, real GDP growth (NGDP_RPCH), fetched via DBnomics because imf.org blocks automated downloads.

api.worldbank.org/v2 · api.db.nomics.world (IMF/WEO:2019-10) · retrieved Nov 2026

SOURCE 4 · OXFORD COVID-19 GOVERNMENT RESPONSE TRACKER

What governments did, day by day

National policy data from 1 January 2020 to 31 December 2022: every containment, economic, health and vaccination indicator, plus the indexes.

The file is 44 MB, so it is downloaded by the build script and kept out of git.

Two things we take from it

Stringency index (0–100), via OWID: how strict the measures were → *freedom*

C1M school closing (0–3), straight from Oxford: level 3 = closed at all levels → *education*

github.com/OxCGRT/covid-policy-dataset · Blavatnik School of Government · retrieved 2026-11-03

PART 3 · HOW WE PROCESSED IT

Five stages from raw files to the app

1

Raw sources

CSV and API downloads

2

covid-combo.db

SQLite, 175 MB: raw tables plus derived tables

3

Parquet export

0.9 MB in git: select and filter only

4

resilience.py

One row per country, scores and ranks

5

app.py

Streamlit UI, ten tabs

Fixed calculations (GDP loss, school weeks, unemployment rise…) happen once, in **build_combo_db.py**.

Everything that depends on the sidebar (end date, baseline, peers, weights) happens live, in **resilience.py**.

Rebuild: python build_combo_db.py (--refresh re-downloads) → python export_app_data.py

DERIVED · ECONOMY & RECOVERY

GDP loss: actual GDP vs. a world without COVID

IMF baseline (default)

Grow actual 2019 GDP with the IMF's Oct 2019 forecast for each year 2020–23.

F(y) = GDP2019 × Π (1 + g/100)

Trend baseline

Extend the average growth of 2015–19 over 2020–23.

g = (GDP2019 / GDP2015)^(1/4) – 1

Loss % = (Σ expected – Σ actual) ÷ Σ expected, summed over 2020–2023

Recovery = number of years in 2020–2024 that real GDP stayed below its 2019 level (0–5).

Table gdp_covid_loss · gdp_loss_for_country(), imf_baseline_loss(), gdp_recovery() in build_combo_db.py

DERIVED · JOBS, EDUCATION, LIFE EXPECTANCY

Yearly and daily series become one number each

Criterion	Calculation	From
School weeks closed	Days with C1M = 3 in 2020–22, ÷ 7	OxCGRT daily
Unemployment rise	Highest rate 2020–22 − 2019, in %-points	World Bank yearly
Life-expectancy drop	2019 − lowest of 2020–22, in years	World Bank yearly
Health capacity	Latest value 2015–19 per indicator	World Bank yearly

Example, unemployment rise: Belgium +0.88 (temporary unemployment kept jobs formal) · USA +4.39 (mass lay-offs in 2020) · Italy −0.39 (people stopped looking for work, so a negative value is not good news)

Tables society_covid, health_capacity · school_closure_weeks(), social_changes(), health_capacity() · pandas pivot, no loops

LIVE IN THE APP · RESILIENCE.PY

Daily data, cut at the end date

Metric	How it is taken from the daily series
Excess deaths / M	Last cumulative value on or before the end date; empty if older than 6 months
Reported deaths / M	Highest total_deaths_per_million up to the end date
Avg. stringency	Mean stringency_index from 2020 to the end date (at most 2022)
Days to 50% vaccinated	First date ≥ 50% fully vaccinated, minus 8 Dec 2020
Excess vs. reported	Excess ÷ reported deaths: a data-quality signal

Then joined on ISO code with GDP loss, society and health capacity, and filtered to the chosen peers (minimum population, continents, countries).

country_metrics(cutoff) · with_gdp_baseline() · select_peers() · end dates: end of 2020, 2021, 2022 or 2023

FROM CRITERIA TO A RANKING

Rank, weight, then test the weights

1

Rank per criterion

Each criterion becomes a rank scaled 0 (best) to 1 (worst), so deaths and weeks are comparable.

2

Weighted score

Score = 100 × (1 – weighted badness). 100 = best on every criterion. Countries missing a criterion are listed, not hidden.

3

Stability check

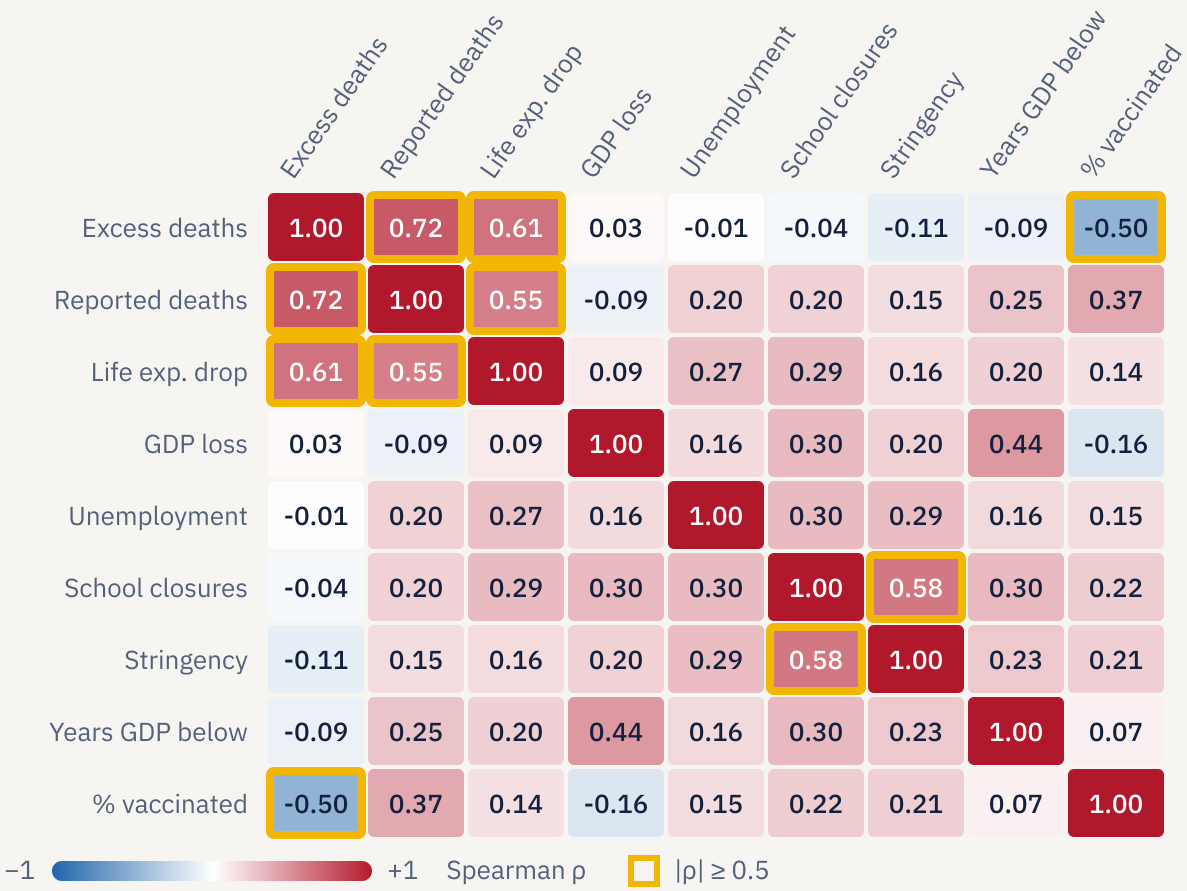
Re-rank under every weight combination on a 0.1 grid. Top in all of them = a robust result.

Default weights: **Health 5** · Economy 2 · Jobs 1 · Education 1 · Freedom 1 · Recovery 1

score_countries() · weight_grid() · rank_stability() · capacity_correlations() · measure_correlations() · strongest_pairs()

WHAT RELATES TO WHAT

Correlations between all measures



What it shows

Freedom vs. deaths: stringency and excess deaths correlate at ρ = -0.11 (67 countries), so strict rules did not go with fewer or more excess deaths.

Vaccination: a higher share fully vaccinated goes with fewer excess deaths (ρ = -0.50).

Framed pairs (|ρ| ≥ 0.5): excess vs. reported deaths (0.72) and life-expectancy drop (0.61), reported deaths vs. life-expectancy drop (0.55), and school closures vs. stringency (0.58), so the rules cluster.

Careful: many measures track national wealth, and correlation is not causation.

Correlations tab in the app · measure_correlations(), strongest_pairs() · all countries ≥ 1M people, end 2023, IMF baseline

LIMITS OF THE DATA

What to keep in mind

Excess mortality coverage

Available for about 125 countries; it is not age-standardised, so older populations look worse.

GDP loss is not only COVID

The 2022 energy crisis, wars and oil booms all land in the same gap.

Policy data stops in 2022

Stringency and school closures cover 2020–2022 only, whatever the end date.

Codes differ per source

OWID, World Bank and IMF disagree on some ISO codes (Kosovo) and on what counts as a country.

The app’s Data quality tab shows which countries miss which criterion

In short

Goal: rank countries on how well they came through COVID-19, with resilience defined as six explicit criteria.

Data: OWID, Oxford OxCGRT, the World Bank and the IMF's pre-COVID forecast, all public and dated.

Processing: one SQLite database, derived tables for each criterion, a small Parquet export and a live, weight-tested score.

Questions? Try it live: https://covid-resilience.streamlit.app/

Or run it locally: streamlit run app.py

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